SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

61.501 Production Machine Learning, 2026
Homework 1

Due 18 Oct 2026, 11:59 pm

This homework will be graded by TA Zhuochen Wu

1. Linear classification [15 points]

Automatic handwritten digit recognition is an important machine learning task. The US Postal Service Zip Code Database (http://www.unitedstateszipcodes.org/zip-code-database/) provides 16 $\times$ 16 pixel images preprocessed from scanned handwritten zip codes (US zip codes are the analogues of Singapore postal codes). The task is to recognize the digit in each image. We shall consider the simpler goal of recognizing only two digits: 1 and 5. To simplify our task even further, let's consider only two features: intensity and symmetry. Digit 5 generally occupies more black pixels and thus have higher average pixel intensity than digit 1. Digit 1 is usually symmetric but digit 5 is not. By defining asymmetry as the average difference between an image and its flipped versions, and symmetry as the negation of asymmetry, we can get higher symmetry values for digit 1.

Write an implementation of the perceptron algorithm. Train it on the training set (`train 1 5.csv`), and evaluate its accuracy on the test set (`test 1 5.csv`). The training and test sets are posted on eDimension. csv stands for comma-separated values. In the files, each row is an example. The first value is the symmetry, the second is the average intensity, and the third is the label.

Note: please do NOT shuffle the data. Visit the instances sequentially in the training set when running the perceptron algorithm.

(a) [5 points] Run the perceptron algorithm with offset for 1 epoch (i.e., traversing the training set 1 time), report the θ, offset and accuracy on the test set.

(b) [5 points] Run the perceptron algorithm with offset for 5 epochs, report the θ, offset and accuracy on the test set.

(c) [5 points] Submit your code for parts (a) and (b) in clearly organized manner. Ensure that the code is executable so that the TA can run it and and reproduce expected results.

2. Linear and polynomial regression [25 points]

For this exercise, you will experiment with linear and polynomial (in features) regression on a given data set. The inputs are in the file `hw1x.dat` and the desired outputs in `hw1y.dat`.

(a) [10 points] Load the data and add a column vector of 1s to the inputs, write a function implementing the closed form linear regression formula discussed in class to obtain the weight vector θ and report it. Plot both the linear regression line and the data on the same graph. Write a function that will evaluate the training error in terms of empirical risk of the resulting fit and report the error.

(b) [5 points] Write a function to calculate the weight vector θ using gradient descent algorithm. Consider learning rate as $\eta = 0.01$ and run it for 5 epochs. Report θ and training error for the minimum empirical risk. Repeat the same with stochastic gradient descent algorithm.

(c) [10 points] Write a function called PolyRegress(x,y,d) which adds the features $x^2, x^3, ...x^d$ to the inputs and performs polynomial regression using closed form solution. Use your function to get a quadratic fit of the data. Plot the data and the fit. Report the training error. Repeat the same for 3rd order fit to 15th order fit. After which order fit does the error get worse?

3. Ridge regression [20 points]

In this problem, we will explore the effects of ridge regression on generalization. We will use `hw1_ridge_x.dat` as the inputs and `hw1_ridge_y.dat` as the desired output. Please note that a column vector of 1s is already added to the inputs. Recall from Lecture Notes Week2_notes01-LinearRegression, the optimal weight for ridge regression is given by

$$\hat{\theta} = (n\lambda I + X^T X)^{-1} X^T Y \tag{1}$$

To find a suitable value for λ, we will set aside a small subset of the provided data set for estimating the test loss. This subset is called *validation set*, which we use to compute *validation loss*. The remainder of the data will be called the *training set*. Let the first 10 entries of the data set be the validation set, and the last 40 entries be the training set. Concatenate their features into matrices vX and tX, and their responses into vectors vY and tY.

(a) [10 points] Write a function *ridge_regression*(tX, tY, l) that takes the training features, training responses and regularizing parameter λ, and outputs the exact solution θ for ridge regression. Report the resulting value of θ for $\lambda = 0.15$.

(b) [10 points] Use the following code to plot graphs of the validation loss and training loss as λ varies on logarithmic scale from $\lambda = 10^{-5}$ to $\lambda = 10^{0}$. Write down the value of λ that minimizes the validation loss.

```
import matplotlib.pyplot as plt

tn = tX.shape[0]
vn = vX.shape[0]
tloss = []
vloss = []
index = -np.arange(0,5,0.1)

for i in index:
    w = ridge_regression(tX,tY,10**i)
    tloss = tloss+[np.sum((np.dot(tX,w)-tY)**2)/tn/2]
    vloss = vloss+[np.sum((np.dot(vX,w)-vY)**2)/vn/2]
```

```
plt.plot(index,np.log(tloss),'r')
plt.plot(index,np.log(vloss),'b')
```

4. Support Vector Machines [20 points]

(a) Understanding SVM Concepts [5 points]

Explain the concept of the maximum margin separator and how it relates to Support Vector Machines. In your explanation, include the following:

- The role of the hyperplane in SVM.
- Incorporation of regularization into SVM formulation to handle non-linearly separable data.
- Discuss the potential implications and consequences if, hypothetically, we were to minimize the margin instead of maximizing it. What would be the effect on the classifier's performance and generalization ability?

(b) Understanding SVM with Soft Margin [5 points]

Support Vector Machines (SVM) with a soft margin allow for some misclassifications to handle non-linearly separable data more effectively. Consider the following SVM formulation with a soft margin:

Primal problem:

$$\min_{\mathbf{w},b,\xi} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n}\xi_i$$

subject to

$$y_i(\mathbf{w}\cdot\mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad i = 1,\ldots,n$$

- Describe how the introduction of slack variables (ξi) affects the constraints and the objective function of the primal problem.
- Given a dataset that is not linearly separable, explain how you would choose the parameter C to balance the trade-off between margin width and classification error.

(c) Understanding Kernel [10 points]

Kernel functions can transform linear classifiers into powerful non-linear classifiers by implicitly mapping data into higher-dimensional spaces. Based on the properties and rules of kernel functions, answer the following questions.

1. **Kernel Construction**: Construct a new kernel function using the following two kernels and verify its validity:

   - $K_1(x, x') = x \cdot x'$
   - $K_2(x, x') = (x \cdot x')^2$

   Combine these kernels using addition and multiplication rules and demonstrate that the resulting functions are valid kernels.

2. **Kernel Verification**: Determine if the following kernel function is valid. Justify your answer using the rules of kernel functions:

   - $K(x, x') = \|x\|^2 + \|x'\|^2 - 2(x \cdot x')$

5 Logistic Regression [20 points]

(a) Logistic Regression Classifier [5 points]

Suppose that you have trained a logistic regression classifier, a nd i t o utputs o n a n ew e xample a prediction $h_\theta(x) = 0.65$. This means (check all that apply):

1. Our estimate for $P(y = 0|x; \theta)$ is 0.35
2. Our estimate for $P(y = 0|x; \theta)$ is 0.65
3. Our estimate for $P(y = 1|x; \theta)$ is 0.35
4. Our estimate for $P(y = 1|x; \theta)$ is 0.65

Please explain your answer.

(b) Logistic Regression Implementation [15 points]

Using the provided dataset provided in folder 5 (455 training samples and 114 test samples), implement a logistic regression classifier from scratch to predict whether a breast-mass sample is benign (0) or malignant (1) from its mean radius and mean texture. Report the test accuracy and display the training loss curve and decision boundary.